

Superstore Profit Analysis

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import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the dataset
df = pd.read_csv('Sample - Superstore.csv', encoding='latin-1')

# 1. DATA PREPARATION (Cleaning and Feature Engineering)
# Convert 'Order Date' to datetime objects
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True, errors='coerce')

# Add time-based columns
df['Order Year'] = df['Order Date'].dt.year
df['Order Quarter'] = df['Order Date'].dt.to_period('Q')
df['Order Month'] = df['Order Date'].dt.to_period('M')

# Calculate Profit Margin
df['Profit Margin'] = df['Profit'] / df['Sales']

# 2. ANALYSIS 1: PRODUCT PERFORMANCE
print("--- 1a. Performance by Product Category (Sorted by Profit) ---")
category_performance = df.groupby('Category').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()
category_performance['Profit_Margin'] = category_performance['Total_Profit'] / category_performance['Total_Sales']
category_performance = category_performance.sort_values(by='Total_Profit', ascending=False).round(2)
print(category_performance.to_markdown(index=False, numalign='left', stralign='left'))

print("\n--- 1b. Performance by Product Sub-Category (Sorted by Profit) ---")
sub_category_performance = df.groupby('Sub-Category').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()
sub_category_performance['Profit_Margin'] = sub_category_performance['Total_Profit'] / sub_category_performance['Total_Sales']
sub_category_performance = sub_category_performance.sort_values(by='Total_Profit', ascending=False).round(2)
print(sub_category_performance.to_markdown(index=False, numalign='left', stralign='left'))

# 3. ANALYSIS 2: CUSTOMER SEGMENT PERFORMANCE
print("\n--- 2. Performance by Customer Segment (Sorted by Profit) ---")
segment_performance = df.groupby('Segment').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()
segment_performance['Profit_Margin'] = segment_performance['Total_Profit'] / segment_performance['Total_Sales']
segment_performance = segment_performance.sort_values(by='Total_Profit', ascending=False).round(2)
print(segment_performance.to_markdown(index=False, numalign='left', stralign='left'))

# 4. ANALYSIS 3: TIME SERIES / SEASONALITY (Quarterly)

# Aggregate Sales and Profit by Quarter
quarterly_performance = df.groupby('Order Quarter').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum')
).reset_index()
quarterly_performance['Quarter'] = quarterly_performance['Order Quarter'].astype(str)

# Plotting the Quarterly Sales Trend
plt.figure(figsize=(12, 6))
sns.lineplot(x='Quarter', y='Total_Sales', data=quarterly_performance, marker='o', label='Total Sales')
sns.lineplot(x='Quarter', y='Total_Profit', data=quarterly_performance, marker='o', label='Total Profit')
plt.title('Quarterly Sales and Profit Trend (2014-2017)', fontsize=14)
plt.xlabel('Order Quarter', fontsize=12)
plt.ylabel('Amount (USD)', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend()
plt.tight_layout()
plt.savefig('quarterly_sales_profit_trend.png')
plt.close()

print("\n--- 3. Quarterly Sales and Profit Performance (Plot saved as quarterly_sales_profit_trend.png) ---")
print(quarterly_performance[['Quarter', 'Total_Sales', 'Total_Profit']].round(2).to_markdown(index=False, numalign='left', s

# 5. ANALYSIS 4: GEOGRAPHIC PERFORMANCE

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print("\n--- 4a. Performance by Region (Sorted by Profit) ---")
region_performance = df.groupby('Region').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()
region_performance['Profit_Margin'] = region_performance['Total_Profit'] / region_performance['Total_Sales']
region_performance = region_performance.sort_values(by='Total_Profit', ascending=False).round(2)
print(region_performance.to_markdown(index=False, numalign='left', stralign='left'))

print("\n--- 4b. Top 5 Least Profitable Cities (Loss Drivers) ---")
city_performance = df.groupby('City').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()
bottom_5_cities = city_performance.sort_values(by='Total_Profit', ascending=True).head(5)
bottom_5_cities[['Total_Sales', 'Total_Profit']] = bottom_5_cities[['Total_Sales', 'Total_Profit']].round(2)
print(bottom_5_cities.to_markdown(index=False, numalign='left', stralign='left'))

# 6. ANALYSIS 5: DISCOUNT IMPACT

# Create discount bins
bins = [-0.01, 0.001, 0.201, 0.401, 0.601, 1.01]
labels = ['0%', '1%-20%', '21%-40%', '41%-60%', '61%+' ]
df['Discount_Bin'] = pd.cut(df['Discount'], bins=bins, labels=labels, right=False)

discount_analysis = df.groupby('Discount_Bin').agg(
    Total_Sales=('Sales', 'sum'),
    Total_Profit=('Profit', 'sum'),
).reset_index()

discount_analysis['Profit_Margin'] = discount_analysis['Total_Profit'] / discount_analysis['Total_Sales']
discount_analysis = discount_analysis.sort_values(by='Discount_Bin', ascending=True).round(2)

# Plotting the Profit Margin by Discount Bin
plt.figure(figsize=(10, 6))
sns.barplot(x='Discount_Bin', y='Profit_Margin', data=discount_analysis, palette='viridis')
plt.title('Profit Margin by Discount Level', fontsize=14)
plt.xlabel('Discount Level', fontsize=12)
plt.ylabel('Average Profit Margin', fontsize=12)
plt.yticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.savefig('profit_margin_by_discount.png')
plt.close()

print("\n--- 5. Impact of Discount on Profitability (Plot saved as profit_margin_by_discount.png) ---")
print(discount_analysis.to_markdown(index=False, numalign='left', stralign='left'))

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--- 1a. Performance by Product Category (Sorted by Profit) ---
| Category      | Total_Sales | Total_Profit | Profit_Margin |
|:-----|:-----|:-----|:-----|
| Technology    | 836154      | 145456       | 0.17          |
| Office Supplies | 719047      | 122491       | 0.17          |
| Furniture     | 742000      | 18451.2      | 0.02          |

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--- 1b. Performance by Product Sub-Category (Sorted by Profit) ---
| Sub-Category  | Total_Sales | Total_Profit | Profit_Margin |
|:-----|:-----|:-----|:-----|
| Copiers       | 149528      | 55617.9      | 0.37          |
| Phones        | 330007      | 44516.2      | 0.13          |
| Accessories    | 167380      | 41936.8      | 0.25          |
| Paper         | 78479.2     | 34053.3      | 0.43          |
| Binders       | 203413      | 30221.6      | 0.15          |
| Chairs        | 328449      | 26590.2      | 0.08          |
| Storage       | 223844      | 21279        | 0.1           |
| Appliances    | 107532      | 18138.1      | 0.17          |
| Furnishings   | 91705.1     | 13059.2      | 0.14          |
| Envelopes     | 16476.4     | 6964.1       | 0.42          |
| Art           | 27118.8     | 6527.96      | 0.24          |
| Labels        | 12486.3     | 5546.18      | 0.44          |
| Machines      | 189239      | 3384.73      | 0.02          |
| Fasteners     | 3024.25     | 949.53       | 0.31          |
| Supplies      | 46673.5     | -1188.99     | -0.03         |
| Bookcases     | 114880      | -3472.56     | -0.03         |
| Tables        | 206966      | -17725.6     | -0.09         |

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--- 2. Performance by Customer Segment (Sorted by Profit) ---
| Segment      | Total_Sales | Total_Profit | Profit_Margin |
|:-----|:-----|:-----|:-----|
| Consumer     | 1.1614e+06  | 134119       | 0.12          |
| Corporate    | 706146      | 91979.4      | 0.13          |
| Home Office  | 429653      | 60299        | 0.14          |

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--- 3. Quarterly Sales and Profit Performance (Plot saved as quarterly_sales_profit_trend.png) ---
| Quarter      | Total_Sales | Total_Profit |

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:-----	:-----	:-----	
2014Q1	38633.7	5375.39	
2014Q2	39663.8	7279.8	
2014Q3	54179.1	2891.79	
2014Q4	46430.1	3444.9	
2015Q1	42927.4	3676.46	
2015Q2	35559.8	3534.13	
2015Q3	50991.8	9155.24	
2015Q4	34734.5	3349.33	
2016Q1	95694.7	22467.9	
2016Q2	60691.2	9689.57	
2016Q3	50517.3	4978.72	
2016Q4	64586.6	7388.97	
2017Q1	95727.8	12018.6	
2017Q2	50074.8	1358.2	
2017Q3	75083.1	8753.19	
2017Q4	52421.5	3441.44	

--- 4a. Performance by Region (Sorted by Profit) ---

Region	Total Sales	Total Profit	Profit Margin	
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